

Sellers' Patience and Bargaining Outcomes: Evidence from the U.S. Real Estate Market

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Abstract In the process of real estate transactions, the final sales price of a property is influenced by numerous factors. This study, based on a dataset from the U.S. real estate market, employs regression analysis to explore the potential relationship between sellers' patience and bargaining outcomes. Furthermore, it discusses the possibility of a causal relationship between these variables. This report presents a detailed account of each task from Task 1 to Task 6, including the data processing procedures, analytical methods, reasons for method selection, and corresponding results. In addition, several important supporting documents are attached to this report.

Keywords Machine Learning, Sentiment Analysis, Regression Analysis, Endogeneity, High-Dimensional Fixed Effects Model, Instrumental Variables, Heckman Selection Model

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Task 1: Data Coverage Map

The primary objective of this task is to generate a U.S. map illustrating the geographical distribution of real estate listings, specifically focusing on the number of bargaining events occurring in each state as recorded in the dataset. The map utilizes a gradient color scheme to represent the frequency of these events across states, enabling a visual analysis of regional patterns.

This task involves two core steps:

- a. Identification of the state where each property is located.
- b. Detection of bargaining events and their aggregation by state.

To determine the location of each listing, I used the variables **zipcode** and **fips** from the dataset. The zipcode corresponds to the United States postal code system, while fips represents the Federal Information Processing Standard code, both of which can be used to identify the state and county of a listing.

I first obtained a publicly available mapping table that links zip codes to U.S. states. Using Excel's VLOOKUP function, I matched the zipcode of each listing to its corresponding state and created a new variable, **state**. However, some zip codes in the dataset appeared to contain only four digits instead of the standard five, likely due to formatting errors or missing data. As a result, 41 listings could not be matched. To address this, I used the fips variable in combination with a FIPS-to-state reference table to supplement and complete the state information for these records.

After all listings were successfully assigned to a state, I defined a "bargaining event" as a case where the initial listing price (**initial_list_price**) differed from the final sales price (**sales_price**), provided that the sales_price was not missing. Based on this definition, I computed the number of bargaining events in each state.

Using Python, I visualized these statistics by generating both static and interactive versions of a U.S. map. In these maps, darker shades indicate a higher number of bargaining events, offering an intuitive view of the dataset's geographical distribution and density. Both the dynamic interactive map (stored in **bargaining_distribution_US_interactive.html**) and the static map (stored in **bargaining_distribution_US_static.png**) are attached to the report as supplementary files.

The map reveals that bargaining events are predominantly concentrated in coastal areas, with California having the highest number, followed closely by Florida.

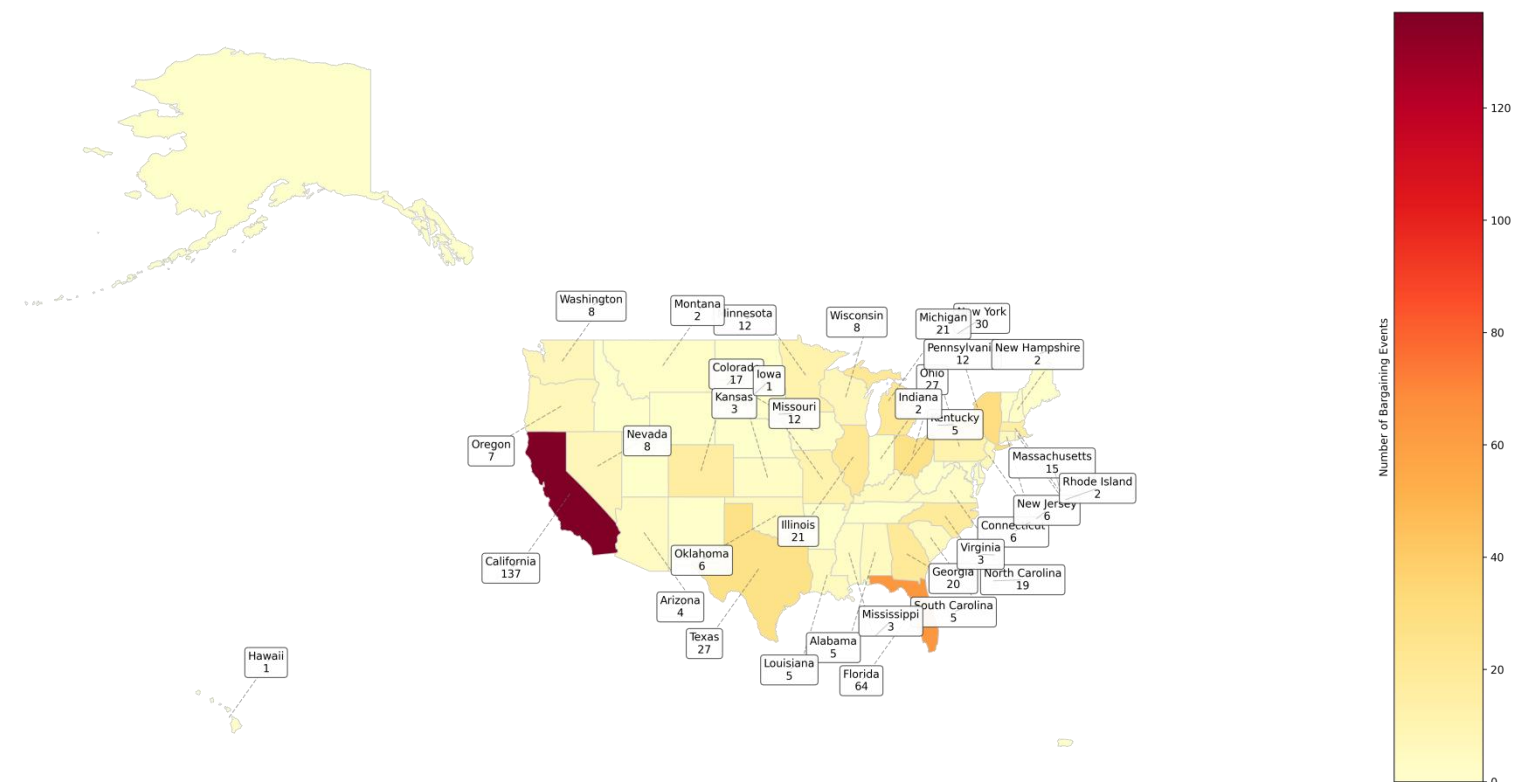


FIGURE 1. Geographical Distribution of Bargaining Events in U.S. Real Estate Listings

Task 2: Data Cleaning and Preprocessing

In the second task of this study, I conducted systematic data cleaning and preprocessing of the dataset, laying a solid foundation for subsequent text analysis and regression analysis. The data cleaning process focused primarily on numerical variables, non-numerical variables, and time variables, with special attention to missing values, outliers, and data formatting issues.

1. Cleaning of Numerical Variables and Handling of Missing Values

Before beginning the cleaning process, I first examined the overall data structure and summarized the missingness of each variable. I found that the following important variables exhibited varying degrees of missingness: `lot_size_acres`, `sales_price`, `bedrooms`, `baths`, `stories`, `living_area_ft`, `dom`, and `prop_age`.

1.1 Processing of `sales_price` and `dom`

Since there are both sold and unsold properties in the dataset, the absence of `sales_price` and `dom` for unsold listings is considered reasonable and should be preserved. However, during inspection, I discovered that some properties had a recorded `sales_price` but were missing the `dom` value, which likely resulted from data

entry errors. Based on logical reasoning, any property with a sales price should have a recorded number of days on market. Therefore, I removed these inconsistent observations, resulting in the deletion of 32 records.

1.2 Handling Missing Values in Property Attribute Variables

1.2.1 Processing of lot_size_acres

This variable has 49 missing values, a relatively low proportion, and imputation is feasible. In the U.S. real estate market, not all property types explicitly list lot_size_acres, especially condominiums (**condos**) and cooperatives (**co-ops**), where the lot size is usually shared by the entire building, making the missingness justifiable.

I examined the property type (prop_type) of the missing values and found that the vast majority of missing cases were concentrated in prop_type = 3/4 (condo or co-op), with all entries of prop_type = 4 missing lot_size_acres, confirming the reasonable nature of the missingness.

Thus, for observations where prop_type equals 3 or 4, I retained the missing values; for prop_type equal to 1, 2, or 5, I filled missing values using the group mean within each prop_type to ensure the imputed values were reasonable and preserved heterogeneity, as different prop_types may vary substantially in lot size. Therefore, using type-specific averages is more appropriate than relying on a global mean.

1.2.2 Processing of bedrooms, baths, stories, living_area_ft, and prop_age

These variables are fundamental housing characteristics and should follow basic logical constraints. For example, bedrooms, baths, and prop_age should not be less than zero, and stories and living_area_ft should be greater than zero. Also, bedrooms and baths should be integers. Therefore, I first removed outliers that violated common-sense boundaries.

Upon inspecting the data structure, I found 3 missing values for bedrooms, 1 for baths, 48 for stories, 19 for living_area_ft, and 12 for prop_age, all of which are relatively small in proportion. I adopted a median imputation strategy grouped by prop_type, to minimize the influence of outliers and heterogeneity across property types. During the imputation of bedrooms and baths, if the imputed value was a decimal, I applied standard rounding.

2. Processing of Non-Numerical Variables

2.1 Condition, Quality, Cooling, Heating, Parking, Pool and Fireplace

These are categorical variables, some of which are ordinal (e.g., condition, quality), while others are binary (e.g., cooling, heating, parking). I uniformly converted empty strings to missing values and standardized the variables as follows:

The missing proportion for condition and quality is relatively high (approximately 50%), but due to their importance in property valuation and transaction processes, I treated missing values as a new category—“Not Disclosed”—and assigned them a value of 0. Other categories were assigned values in descending order, from the highest to the lowest level of evaluation.

For binary variables such as cooling, heating, and parking, I used YES = 1 and NO = 0.

Due to the extremely high missing rates for pool and fireplace, these variables were excluded from further regression analysis.

2.2 Preprocessing of Text Variable publicremarks

The publicremarks variable contains property description text, which will be used for text analysis in Task 4. In this task, I conducted preliminary cleaning, including case normalization and punctuation removal, to prepare the text for later analysis.

3. Processing of Time Variables

The dataset contains the variable **initial_list_date**, which records the listing date of the property. I converted this variable into a Stata-recognizable standard date format (date) to facilitate its use in subsequent regression analysis.

Task 3: Summary Statistics

After completing the data cleaning process, I further conducted a descriptive statistical analysis of the dataset to understand the distribution of key variables in real estate transactions and to provide necessary background information and data characteristics for the subsequent regression analysis (Task 5). In this task, the unit of analysis is at the listing level.

The variables analyzed in this section are divided into two main categories: variables related to bargaining outcomes and key property characteristics.

1. Variable Description

1.1 Variables Related to Bargaining Outcomes

The variables directly related to bargaining outcomes include **initial_list_price**, **sales_price**, and **dom**.

In addition, I defined a new variable **discount**:

$$discount = \frac{initial_list_price - sales_price}{initial_list_price}$$

The variable discount represents the discount rate and is used to quantify the degree of price change during the bargaining process. This variable is a key indicator of bargaining outcomes and serves as the **dependent variable** in the subsequent regression analysis.

1.2 Key Property Characteristic Variables

The dataset contains a large number of variables related to property characteristics. These include: prop_age, prop_type, living_area_ft, bedrooms, baths, stories, condition, quality, cooling, heating, parking, pool, and fireplace. As discussed in previous sections, some of these variables are categorical variables, while others are numerical variables. These variables encompass aspects such as housing quality and facility configurations, and may serve as control variables in subsequent tasks.

2. Descriptive Statistics and Result Presentation

I used the estpost tabstat command in Stata to compute summary statistics for the above variables. The statistical measures include mean, standard deviation, minimum, median, and maximum.

The results have been exported to two files, **summary_categoricalvariables.csv** and **summary_numericalvariables.csv**, both of which are attached to the report as supplementary files.

Below is the presentation of the descriptive statistics results:

TABLE 1. Descriptive Statistics of the Main Numerical Variables

	mean	sd	p50	min	max
discount	0.0480449	0.0961556	0.0335176	-0.494382	0.5664861
initial_list_price	346396.3	294024.1	254900	19000	2119950
sales_price	318476.2	263219.5	243475	21050	1830000
dom	63.97872	76.04521	39	0	477
prop_age	32.74394	28.01161	27	0	176
living_area_ft	1824.577	816.1756	1652	428	5974
bedrooms	3.189158	1.029012	3	1	9
baths	2.268743	0.9258807	2	0	7
stories	1.540946	0.8329227	1	1	12
Observations	867				

Table 2 Descriptive Statistics of the Main Categorical Variables

		count	percentage
cooling	Missing	95	10.95732
	N	42	4.844291
	Y	730	84.19839
	Total	867	100
	N	867	
fireplace	Missing	306	35.29412

	N	94	10.84198
	Y	467	53.8639
	Total	867	100
	N	867	
heating	Missing	37	4.267589
	N	3	.3460208
	Y	827	95.38639
	Total	867	100
	N	867	
parking	Missing	114	13.14879
	N	25	2.883506
	Y	728	83.9677
	Total	867	100
	N	867	
pool	Missing	466	53.74856
	N	166	19.14648
	Y	235	27.10496
	Total	867	100
	N	867	
quality	Missing	503	58.01615
	QAV	206	23.76009
	QBA	6	.6920415
	QEC	2	.2306805
	QEX	19	2.191465
	QFA	34	3.921569
	QGO	57	6.574394
	QLO	1	.1153403
	QLU	3	.3460208
	QPO	1	.1153403
	QVV	35	4.036909
	Total	867	100
	N	867	
condition	AVE	241	27.797
	EXC	13	1.499423
	FAI	26	2.998847
	GOO	89	10.26528
	Missing	482	55.594
	POO	5	.5767013
	UCN	1	.1153403
	VGO	10	1.153403
	Total	867	100
	N	867	
prop_type	1	701	80.85352
	2	34	3.921569

3	102	11.76471
4	2	.2306805
5	28	3.229527
Total	867	100
N	867	

Task 4: Sentiment Analysis of Textual Descriptions

Using Machine Learning

In this task, I employed machine learning methods to analyze the textual descriptions of properties (**public remarks**) provided by sellers in real estate listings, with the aim of identifying those who may be eager to sell. In this context, I constructed a variable named *urgent* to capture the seller’s urgency to sell, which will be used to support subsequent regression analysis.

Specifically, I first extracted a sample of 200 listing descriptions from the dataset and manually labeled them. During the labeling process, I examined the vocabulary, tone, and urgency conveyed in the text (e.g., expressions such as “immediate,” “quick,” “motivated”) to determine whether the seller exhibited a strong inclination toward an urgent sale.

Next, I performed text preprocessing on these 200 labeled samples, including stop-word removal, elimination of extra whitespace, and removal of numerical values. I then split the dataset into a training set and a test set at an 8:2 ratio and used the TF-IDF algorithm to vectorize the text for model training and optimization.

For model selection, I compared four commonly used text classification algorithms: Logistic Regression, Random Forest, K-Nearest Neighbors (KNN), and Naive Bayes. By evaluating model performance on the test set and comparing precision, recall, F1-score, and accuracy, I selected the most suitable model.

The performances of the above four models are as follows:

TABLE 3. Performance Comparison Across Four Models

model	Logistic Regression	Random Forest	KNN	Naive Bayes
precision	0.59	0.83	0.61	0.57
recall	0.65	0.75	0.7	0.65
f1-score	0.62	0.79	0.65	0.6
accuracy	0.6	0.8	0.62	0.57

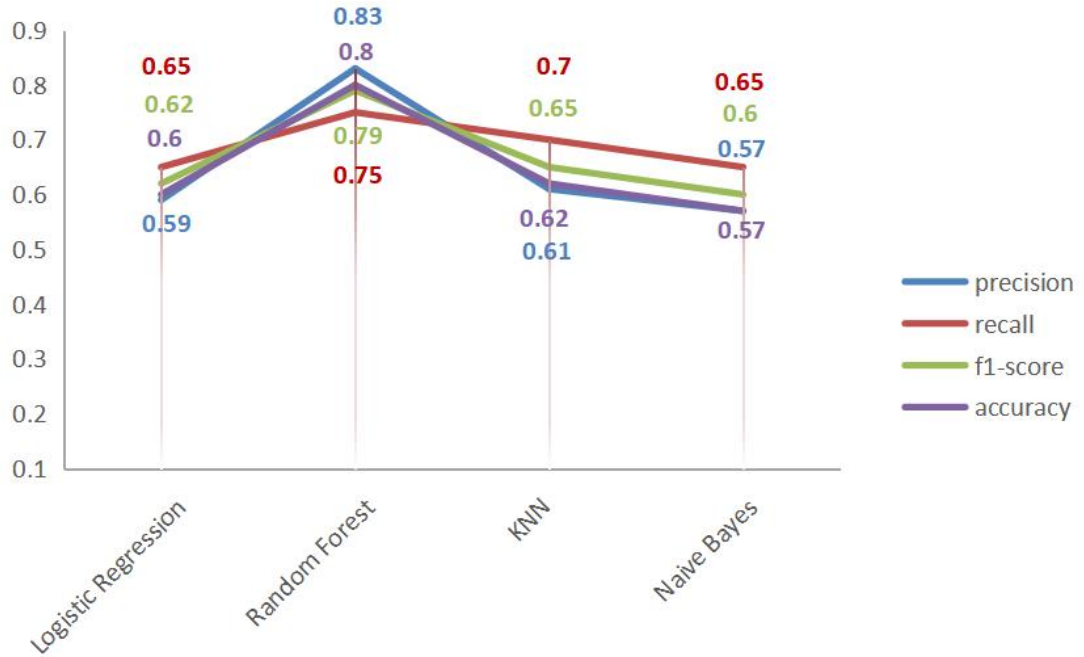


FIGURE 2. Key Metrics Comparison Across the Four Models

The performance comparison indicated that the Random Forest model exhibited strong applicability. Its performance in terms of precision, F1-score, accuracy, and recall significantly outperformed the other models. With a precision score of 0.83, the model exhibited superior performance. In this study, since the TF-IDF transformation produced high-dimensional features, Random Forest, which is particularly well-suited for handling high-dimensional and complex data, performed effectively in feature selection. Moreover, Random Forest's use of random sampling and feature selection helps reduce overfitting during the training process. Therefore, selecting the Random Forest model is both reasonable and effective for this study.

Then I applied the trained model to the remaining unlabeled textual data in the dataset to predict urgency and constructed the new binary variable urgent. Listings predicted to be urgent sales were assigned a value of 1, and others were assigned 0. This variable was later included in the regression model as a control variable reflecting seller characteristics.

The identification results of sellers' urgency sentiment are stored in the file **data_identified.csv**, which is attached to this report. A preliminary statistical summary is presented in Table 4.

TABLE 4. Seller urgency

		count	percentage
urgent	0	550	0.634371396
	1	317	0.365628604
	Total	867	

Task 5: Regression Analysis

In this section, I conduct a regression analysis based on the cleaned and constructed dataset to explore the relationship between sellers' patience and bargaining outcomes.

1. Variable Selection

1.1 Independent and Dependent Variables

First, in terms of variable selection, I use the **dom** as the core independent variable to measure the seller's level of patience. A higher value of dom indicates that the seller is more patient to sell the property.

The dependent variable is the **discount** (defined in **Task 3**), which is calculated as follows:

$$discount = \frac{initial_list_price - sales_price}{initial_list_price}$$

This variable reflects the price discount ratio between the listing price and the final sales price, serving as a measure of bargaining outcomes. A value of discount > 0 indicates that the seller lowered the price; discount < 0 implies that the transaction occurred at a premium. The greater the absolute value of the discount, the larger the deviation of the final price from the initial listing price, suggesting a more intense bargaining process.

1.2 Control Variable Selection

To examine the effect of dom on discount while accounting for other possible confounding factors, a set of control variables is included in the model.

The control variables in this study are considered from the following three dimensions: **property characteristics**, **seller characteristics**, and **macroeconomic factors**.

1.2.1 Property Characteristics

Property-specific variables such as quality, property age, number of stories, number of bedrooms, and the presence of various facilities (e.g. parking, cooling, heating) may influence the bargaining process during a sale.

This category includes: **condition**, **quality**, **cooling**, **heating**, **parking**, **bedrooms**, **prop_age**, **stories**, and **lot_size_acres**. These factors may affect how desirable the property is, thereby influencing both sale price and time on market.

Categorical variables such as **condition**, **quality**, **cooling**, **heating**, and **parking** were converted into numerical variables during the earlier data cleaning process (see **Task 2** for details on the encoding methods). Due to a high proportion of missing values, the variables **pool** and **fireplace** are excluded. Additionally, **prop_type**, being a categorical variable that reflects property type, is not included as converting it to a numerical format would not be meaningful.

1.2.2 Seller Characteristics

Beyond the property itself, a seller's expressed sentiment in the listing may also significantly influence the bargaining process. Sellers who are more eager to sell may be more inclined to adjust the price to facilitate a quicker transaction.

In **Task 4**, I conducted a sentiment analysis on the text variable `publicremarks` using machine learning methods to identify whether the seller appeared eager to sell. Based on this, I constructed a binary variable **urgent** to capture the urgency of the seller. This variable is included in the regression model as a core control variable.

1.2.3 Macroeconomic Factors

At the macro level, key economic indicators such as household income, employment status, the stability of the economic cycle, and market expectations may significantly influence the behavior and decisions of buyers and sellers in the real estate market. Theoretically, variables such as the risk-free interest rate and employment rate could directly or indirectly impact bargaining outcomes. Therefore, it is reasonable to consider macroeconomic variables as potential controls.

However, due to practical constraints in data availability—including limitations in time, geography, and technical capacity—it is currently not feasible to collect and match related macroeconomic data across different regions of the U.S. Therefore, the study does not include macroeconomic controls in the regression model for the time being. Nonetheless, it is important to emphasize that macro-level variables remain potentially important factors, offering substantial scope for future empirical research.

1.2.4 Avoiding Overcontrol and Endogeneity in Variable Selection

In the dataset, `initial_list_price` represents the property's listing price, which may partially reflect both the underlying value of the property and the seller's pricing strategy or psychological expectations. While it could plausibly influence bargaining outcomes, we chose not to include it as a control variable in the regression analysis for several important reasons.

First, **discount** is constructed directly using `initial_list_price` as a denominator, which introduces few potential issues.

Second, the `initial_list_price` is not an exogenous variable—it is strategically set by the seller based on property characteristics, market conditions, and possibly their own patience. For example, a more patient seller (reflected in longer `dom`) may deliberately set a higher initial listing price to allow room for negotiation, and that

pricing strategy subsequently affects the observed discount. In this sense, initial_list_price lies on the causal path between seller patience and bargaining outcomes. Controlling for it would risk introducing overcontrol bias.

For these reasons, initial_list_price is not included as a control variable in our empirical model.

1.3 Correlation Analysis

Considering the potential issue of high correlations among variables—which could result in serious multicollinearity—I conducted a Pearson correlation coefficient analysis. The results are shown in Table 5.

TABLE 5. Pearson Correlation Coefficient Analysis

varname	discount	dom	urgent	condition_numeric	quality_numeric	cooling_numeric	heating_numeric	parking_numeric	bedrooms	baths	prop_age	stories	living_area_ft	lot_size_acres
discount	1.000	0.48	0.137	-0.037	-0.061	0.054	0.059	-0.100	0.034	-0.024	-0.051	-0.008	0.022	0.062
dom	0.481	1.00	0.069	-0.107	-0.043	-0.023	0.058	-0.045	0.047	0.028	-0.061	-0.013	0.086	0.036
urgent	0.137	0.06	1.000	0.035	0.066	0.028	0.004	-0.009	0.142	0.090	0.019	-0.017	0.085	-0.072
condition_numeric	-0.037	-0.1	0.035	1.000	0.252	0.030	-0.019	0.025	0.053	-0.015	0.066	-0.052	0.027	-0.048
quality_numeric	-0.061	-0.0	0.066	0.252	1.000	0.051	0.049	0.027	0.039	0.107	-0.169	-0.065	0.091	-0.004
cooling_numeric	0.054	-0.0	0.028	0.030	0.051	1.000	0.103	0.031	0.065	0.079	-0.195	-0.034	0.081	0.044
heating_numeric	0.059	0.05	0.004	-0.019	0.049	0.103	1.000	-0.010	0.032	0.018	-0.033	-0.033	0.033	0.002
parking_numeric	-0.100	-0.0	-0.009	0.025	0.027	0.031	-0.010	1.000	0.051	0.115	-0.096	0.036	0.122	0.021
bedrooms	0.034	0.04	0.142	0.053	0.039	0.065	0.032	0.051	1.000	0.559	-0.040	0.066	0.661	-0.086
baths	-0.024	0.02	0.090	-0.015	0.107	0.079	0.018	0.115	0.559	1.000	-0.326	0.223	0.756	0.076
prop_age	-0.051	-0.0	0.019	0.066	-0.169	-0.195	-0.033	-0.096	-0.040	-0.326	1.000	-0.014	-0.258	-0.082
stories	-0.008	-0.0	-0.017	-0.052	-0.065	-0.034	-0.033	0.036	0.066	0.223	-0.014	1.000	0.199	0.057
living_area_ft	0.022	0.08	0.085	0.027	0.091	0.081	0.033	0.122	0.661	0.756	-0.258	0.199	1.000	0.062
lot_size_acres	0.062	0.03	-0.072	-0.048	-0.004	0.044	0.002	0.021	-0.086	0.076	-0.082	0.057	0.062	1.000

It can be observed that some variables are highly correlated, particularly bedrooms and baths (0.56), bedrooms and living_area_ft (0.66), and baths and living_area_ft (0.76). To mitigate the effects of multicollinearity on regression results, I selected bedrooms—a more representative variable that shows stronger correlation with the dependent variable discount—for inclusion in the model.

2. Model Specification

Based on the above analysis, the following regression model is constructed:

$$\begin{aligned}
discount = & \beta_0 + \beta_1 \cdot dom + \beta_2 \cdot urgent + \beta_3 \cdot condition_numeric \\
& + \beta_4 \cdot quality_numeric + \beta_5 \cdot cooling_numeric + \beta_6 \cdot heating_numeric \\
& + \beta_7 \cdot parking_numeric + \beta_8 \cdot bedrooms + \beta_9 \cdot prop_age + \beta_{10} \cdot stories \\
& + \beta_{11} \cdot lot_size_acres
\end{aligned}$$

A multicollinearity test was conducted on the model, and the results are shown in Table 6. All variables had VIF values well below 5, with an average VIF of 1.07, indicating that there are no serious multicollinearity issues.

TABLE 6. Collinearity Test

Variable	VIF	1/VIF
prop_age	1.14	0.873728
quality_numeric	1.13	0.884650
condition_numeric	1.10	0.910663
stories	1.10	0.910981
bedrooms	1.10	0.911450
cooling_numeric	1.07	0.934991
dom	1.05	0.956149
heating_numeric	1.04	0.964017
urgent	1.03	0.970747
parking_numeric	1.03	0.971844
lot_size_acres	1.03	0.974467
Mean VIF	1.07	

To empirically test whether seller patience significantly affects bargaining outcomes, I first estimate an **Ordinary Least Squares (OLS)** regression of discount (price discount rate). The baseline regression results are shown in **column (1) of Table 7:**

TABLE 7. The Regression Results

	OLS (1)	HDFE (2)
	discount	discount
dom	0.001*** (0.000)	0.001*** (0.000)
urgent	0.020** (0.008)	0.021** (0.009)
condition_numeric	0.001 (0.002)	-0.001 (0.002)
quality_numeric	-0.002 (0.001)	-0.000 (0.002)
cooling_numeric	0.016 (0.017)	0.002 (0.019)
heating_numeric	-0.016	-0.006

	(0.059)	(0.060)
parking_numeric	-0.037*	-0.034
	(0.021)	(0.022)
bedrooms	-0.001	-0.004
	(0.004)	(0.005)
prop_age	-0.000	-0.000
	(0.000)	(0.000)
stories	-0.003	-0.002
	(0.006)	(0.007)
lot_size_acres	0.003	0.004
	(0.002)	(0.003)
_cons	0.044	0.065
	(0.064)	(0.066)
N	442	440
r2_a	0.245	0.280
F	14.037	9.312
Standard errors in parentheses		
="* p<0.1	** p<0.05	*** p<0.01"

The regression results indicate that the coefficient on dom is approximately **0.001**, and it is statistically significant at the **1% level**. This suggests that **the longer a property remains listed on the market, the more likely it is to sell at a discounted price, and the size of the discount increases with time on market**. Specifically, each additional day on the market is associated with a 0.001 increase in the discount rate—equivalent to an additional **0.1%** decrease in price. This result is consistent with economic intuition: **the more patient sellers are often more likely to accept a greater price concession to close the deal**. Although the coefficient value may appear small, given the typically large monetary amounts involved in real estate transactions, even a small change in the discount rate can translate into a significant price adjustment. Hence, the result has strong economic relevance.

Furthermore, among the control variables, the seller's **urgency to sell (urgent)** is also a significant factor influencing bargaining outcomes. The coefficient on urgent is significantly positive, suggesting that sellers with a greater sense of urgency are more likely to complete transactions at prices below the listing price. In contrast, the variable parking_numeric has a significantly negative effect on discount, indicating that properties with more parking spaces may have stronger bargaining positions and therefore experience smaller discounts. Other variables, such as stories and lot_size_acres, do not show statistically significant effects, but their inclusion helps to enhance the model's overall explanatory power.

Task 6: Discussion on Endogeneity Issues

Although the OLS model in Task 5 reveals statistically significant correlations—specifically, the variable *dom* significantly affects the discount rate, indicating that the more patient the seller is, the more likely it is that the final transaction price will involve a larger discount compared to the listing price—it must be emphasized that such correlations **cannot be simply interpreted as causal relationships**.

The core issue lies in the potential existence of severe **endogeneity problems** in the model, particularly those arising from **omitted variables, sample selection bias, and reverse causality**.

Although the task description states that a conceptual discussion is sufficient, I also attempted to empirically test some of the methods mentioned below. While these attempts are not yet mature, they may provide support and deeper insights for the theoretical discussion of this issue, and allow for further identification of the causal relationship between seller patience and bargaining outcomes.

1. Omitted Variable Bias

In the regression model established in Task 5, although a large number of housing characteristics and seller sentiment variables have been controlled, potential omitted variable bias may still exist, which could affect the explanatory power of the regression results and the validity of causal inference.

In this study, omitted variables may come from multiple sources:

- a. **From the buyer's perspective**, factors such as payment method (e.g., full payment or not) and whether the buyer is a first-time homebuyer can significantly affect the bargaining process, yet these variables are not observed or controlled;
- b. **From the market and macroeconomic perspective**, factors like the activeness of the real estate market in the region, legal systems, local security environment, and certain macroeconomic conditions (e.g., interest rates, unemployment rates, market expectations) may simultaneously affect bargaining outcomes and transaction prices;
- c. **From the seller's perspective**, the subjective urgency to sell may impact willingness to offer a discount, and different motives for selling may be important determinants of bargaining results.

Omitted variable bias may lead to biased estimation of the coefficient of *dom*, thus resulting in incorrect estimation of the causal relationship between sellers' patience and bargaining outcomes.

1.1 Alleviating Endogeneity via High-Dimensional Fixed Effects Model

To mitigate this issue and enhance causal identification, this study further adopts a **High-Dimensional Fixed Effects Model** to verify the robustness of the model.

Given that many unobservable macro-level and institutional factors may be closely related to both the time on the market and the region where the property is located—for example, different macroeconomic conditions (e.g., interest rate, inflation rate) across years and varying supply-demand structures in different states or cities may jointly affect real estate transactions—it is appropriate to include fixed effects for **year** and **state**. The results are shown in **column (2) of Table 7**.

The regression results show that the coefficient of *dom* remains approximately **0.001** and is still highly significant at the **1%** level. The coefficient estimate is also very close to that of the OLS model. This result enhances the credibility of the causal interpretation, suggesting that even after controlling for macro and institutional differences represented by region and time, the seller's patience is still robustly positively correlated with the discount magnitude.

The **adjusted R²** of the **High-Dimensional Fixed Effects Model** increases from 0.245 in the OLS model to 0.280, indicating improved explanatory power after controlling for more unobservable heterogeneity.

Therefore, combining the results of the two models, it can be concluded that greater seller patience is associated with an increased probability and extent of price concessions in the final transaction. This relationship remains exists even after controlling for year and region, showing strong statistical and economic significance.

1.2 Alleviating Endogeneity via Instrumental Variables

Although the previous analysis has applied the **High-Dimensional Fixed Effects Model** to control for the two key dimensions of **time (year)** and **region (state)**, thereby partially mitigating bias from macroeconomic and institutional environment differences, there may still be unobserved heterogeneity at the individual level of buyers and sellers that is not captured.

To alleviate the endogeneity caused by omitted variables, the use of **Instrumental Variables (IV)** is considered appropriate. This involves identifying a variable that influences *dom* but does not directly affect the discount, and is uncorrelated with the error term in the discount equation. A **Two-Stage Least Squares Model (2SLS)** can be constructed to facilitate more credible causal inference.

In this study, I also attempted to design several instrumental variables, but testing results indicate that none of them serve as ideal instruments, and further exploration is needed.

2. Sample Selection Bias

In addition to the endogeneity issue caused by omitted variables, this study may also face challenges from **sample selection bias**.

It is important to note that our current analytical sample includes only those properties that were successfully sold, while listings that remained unsold during the study period are not included. In the real estate market, successful sale is not a random event. Our previous research only selectively observes listings that resulted in successful transactions, which may lead to **selection bias**.

To address this issue, I applied the **Heckman two-step** method for further analysis.

The results are shown in Table 8:

TABLE 8. Regression Results of the Heckman two-step method

	OLS	STEP 1	STEP 2
	(1)	(2)	(3)
	discount	discount	discount
main			
dom	0.001***	0.001***	0.001***
	(0.000)	(0.000)	(0.000)
urgent	0.020**	0.021**	0.022**
	(0.008)	(0.008)	(0.009)
condition_numeric	0.001	0.002	0.003
	(0.002)	(0.002)	(0.002)
quality_numeric	-0.002	-0.002	-0.002
	(0.001)	(0.001)	(0.001)
cooling_numeric	0.016	0.019	0.023
	(0.017)	(0.017)	(0.016)
heating_numeric	-0.016	-0.017	-0.016
	(0.059)	(0.058)	(0.055)
parking_numeric	-0.037*	-0.038*	-0.039**
	(0.021)	(0.020)	(0.019)
bedrooms	-0.001	-0.001	-0.002
	(0.004)	(0.004)	(0.005)
prop_age	-0.000	-0.000	-0.000
	(0.000)	(0.000)	(0.000)
stories	-0.003	-0.003	-0.000
	(0.006)	(0.006)	(0.006)
lot_size_acres	0.003	0.004	0.004*
	(0.002)	(0.002)	(0.002)
_cons	0.044	0.033	-0.005
	(0.064)	(0.064)	(0.061)
status			
urgent		-0.041	-0.032
		(0.100)	(0.096)
condition_numeric		0.040*	0.030
		(0.022)	(0.021)
bedrooms		-0.034	-0.046
		(0.050)	(0.048)

prop_age		-0.002	0.000
		(0.002)	(0.002)
ln_initial_price		-0.131*	-0.052
		(0.067)	(0.057)
year		0.065***	0.046***
		(0.008)	(0.008)
month		0.019	0.017
		(0.015)	(0.013)
_cons		-128.353*	-91.357***
		**	
		(16.951)	(15.040)
/mills			
lambda		0.019	
		(0.019)	
/			
athrho			1.154***
			(0.131)
lnsigma			-2.308***
			(0.050)
N	442	742	742
r2_a	0.245		
F	14.037		
Standard errors in parentheses			

==* p<0.1		** p<0.05	
		*** p<0.01"	

Column (1) corresponds to the baseline regression model from Task 5, while **columns (2) and (3)** show the regression results of the **Heckman two-step** estimation .

In the first-stage selection model, we estimate the likelihood of transaction using a **Probit** model to identify factors influencing whether a property was sold. Then, in the second-stage outcome model, we introduce the **Inverse Mills Ratio** calculated from the first stage to correct the regression on the discount rate.

By comparing the OLS and Heckman results, we find that even after correcting for sample selection bias, the coefficient of dom remains at **0.001** and is still highly significant at the **1%** level. This result is consistent with the OLS regression, indicating that even after accounting for the sample selection mechanism, the longer the property stays on the market, the greater the discount at the final transaction price, and sellers are more likely to make price concessions during the bargaining process.

This result provides additional support for a causal interpretation.

Moreover, the coefficient of **urgent** is 0.020 in the OLS regression, and slightly increases to 0.021 and 0.022 in the Heckman two-step model, remaining significant at

the **5%** level. This further supports the argument that sellers' urgency plays an important role in determining the final transaction price.

3. Reverse Causality

Apart from endogeneity caused by omitted variables, this study may also face **reverse causality** problems, namely that the bargaining process and outcomes itself may affect the seller's level of patience.

For example, the houses priced too high may fail to attract buyers after being listed, thus extending their time on the market. This implies that dom is not entirely determined by the seller's subjective patience but may also result from **buyer feedback** and **market dynamics**. Furthermore, when sellers observe that their property remains unsold for an extended period, they may revise their pricing strategies, creating a potential **reverse causal relationship** in which bargaining process and outcomes influence the duration of the listing.

As a result, the correlation between sellers' patience and bargaining outcomes may be confounded and cannot be directly interpreted as a causal relationship. This reverse causality problem exacerbates the risk of endogeneity in the model and needs to be further identified and corrected using methods such as **Instrumental Variables**.

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APPENDIX

Several important output files and process code files are provided as attachments to this report, organized into two folders named output and code, respectively. All attachments, along with the report itself, are included in the compressed file **“RA_test_Peijing Han.zip”**.